STAT 380 Final Project Report: Modeling Weekly Retail Sales (Walmart)

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Introduction

Weekly sales are one of the most important measures for retail stores because they affect inventory planning, staffing, and promotions. In real life, sales are influenced by many factors like time of year, holidays, store characteristics, and economic conditions. The goal of this project is to explore patterns in weekly sales and test whether certain factors are statistically associated with higher or lower sales.

I used the Walmart weekly sales dataset because it includes weekly sales along with extra variables like store size and unemployment. The dataset is large enough to show clear trends, but it also has noise and many factors that are not directly measured (like promotions or competition). I focused on three research questions: (1) how sales change by month, (2) whether holiday weeks have higher sales, and (3) whether unemployment is related to weekly sales. I also included a prediction section where I tried to classify whether a week is “high sales” or “low sales” using different models.

Dataset Description

This dataset was created by merging three files from the Walmart Kaggle dataset:

- train.csv: weekly sales data with Store, Dept, Date, Weekly_Sales, and IsHoliday
- stores.csv: store information like store Type and store Size
- features.csv: extra variables like Temperature, Fuel_Price, CPI, and Unemployment

After merging, the final dataset contained 421,570 rows and 16 columns. The main response variable for analysis was Weekly_Sales (continuous). For prediction/classification, I created a binary label called High_Sales, described later.

Data Cleaning and Preparation

Before analysis, I prepared the dataset by doing the following:

- Converted the Date column into a proper date format.
- Extracted Year, Month, and Week from Date since sales patterns depend on time of year.

- Treated IsHoliday as a categorical variable (holiday vs non-holiday).
- Removed rows with missing values in key variables needed for analysis, especially Weekly_Sales and Unemployment.

These steps helped make sure the analysis and comparisons were done on clean data.

Results and Findings

RQ1: Monthly pattern in weekly sales
To answer RQ1, I grouped sales by month and computed the average weekly sales for each month. The results showed a clear seasonal pattern. Sales were highest in November and December, with December having the highest average weekly sales (around $19,426) and November also being high (around $17,535). The lowest month was January (around $14,182). This makes sense because November and December include major shopping periods such as Thanksgiving/Black Friday and Christmas. This also shows that if a store is planning staffing or inventory, the late-year months require more preparation than early-year months.

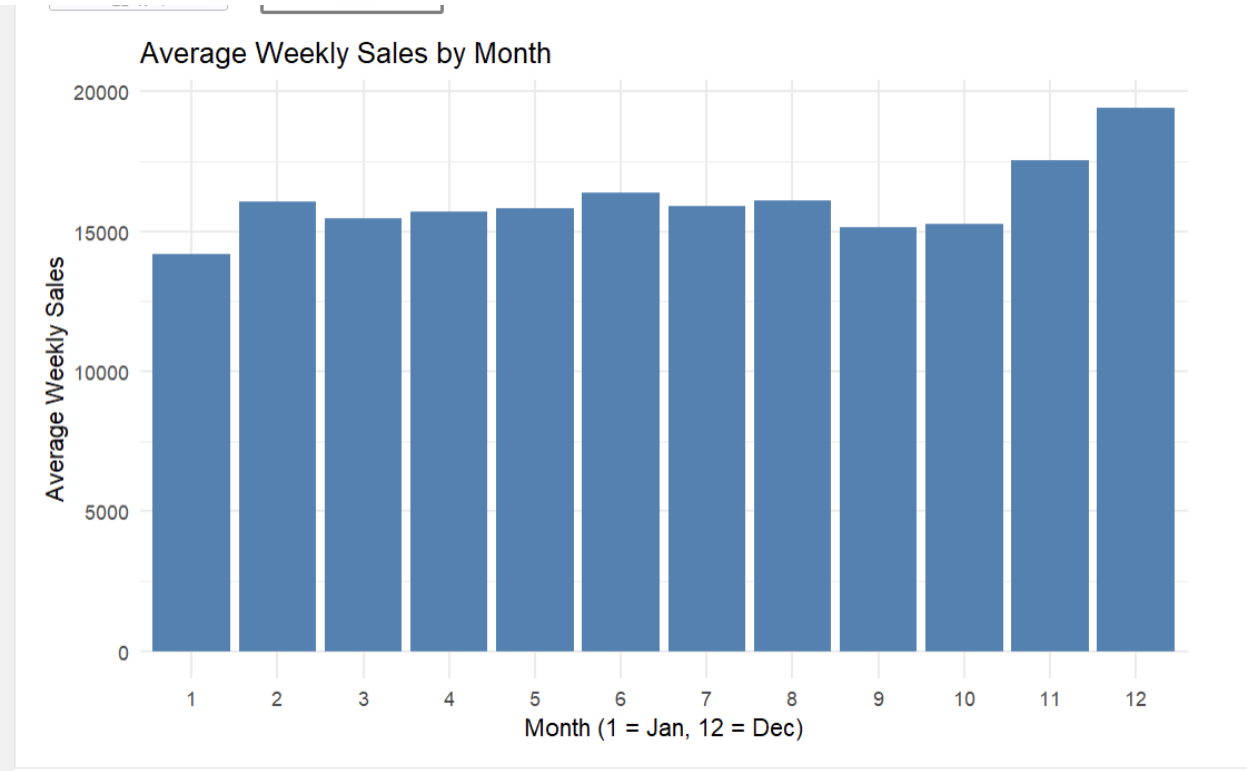

Figure 1: Average Weekly Sales by Month

RQ2: Holiday weeks vs non-holiday weeks
For RQ2, I compared weekly sales between holiday weeks and non-holiday weeks. Holiday

weeks had higher average weekly sales than non-holiday weeks. The non-holiday mean was about $15,952.82 and the holiday mean was about $17,094.30, so holiday weeks were higher by roughly $1,100 on average.

To test whether the difference was statistically meaningful, I ran a two-sample Welch t-test (Welch is appropriate since it does not assume equal variances). The t-test produced t = −7.0255 and p = 2.175e−12, with a 95% confidence interval for (Non-holiday − Holiday) of approximately [−1459.95, −823.02]. Because the p-value is extremely small, this difference is statistically significant, meaning holiday weeks have significantly higher weekly sales on average.

Figure 2: Weekly Sales for Holiday vs Non-Holiday

RQ3: Unemployment and weekly sales

For RQ3, I looked at whether unemployment is connected to weekly sales. First, I computed the correlation between Weekly_Sales and Unemployment, and the correlation was −0.0258, which is extremely close to zero. This suggests there is not a strong linear relationship between unemployment and sales in this dataset.

However, correlation alone can be misleading because unemployment might have a nonlinear effect and because other factors like store size and seasonality could hide the relationship. So I fit a regression model that included unemployment with a quadratic (polynomial degree 2) term and also included other important predictors: IsHoliday, Month, Store, and Size.

A clarification that mattered for this project: even though the model includes a polynomial term for unemployment, it is still linear regression because the model is linear in the coefficients (it is basically using unemployment and unemployment squared as predictors).

In the polynomial model, the unemployment terms were statistically significant at the 0.05 level, but the overall effect was still small compared to other variables. Holiday weeks increased predicted weekly sales by about +$819.8 and Month increased predicted sales by about +$193.3 per month, both capturing strong seasonal and holiday effects. Store size also had a strong positive association with weekly sales. The overall model fit was low, with R-squared around 0.062, which means only about 6.2% of the variation in weekly sales is explained by the model. This suggests weekly sales depend heavily on other factors not included in the dataset, such as promotions, local events, department-specific demand, and competitor behavior.

Figure 3: Weekly Sales vs Unemployment with fitted curve

Prediction / Classification: High vs Low Sales Weeks

Target definition and setup

To turn the dataset into a classification problem, I created a binary variable High_Sales:

- High if Weekly_Sales is above the median
- Low otherwise

I used the median because it provides a simple split into above-typical vs below-typical weekly sales and it is less sensitive to extreme values than using the mean. To avoid class imbalance and make model performance comparisons fair, I created a balanced dataset with 2,000 rows total: 1,000 High and 1,000 Low. I then split the data into training and testing sets. The training set ended up with 702 Low and 698 High, and the test set had 298 Low and 302 High.

Predictors used included Store, Size, Temperature, Fuel_Price, CPI, Unemployment, IsHoliday, Year, and Week.

Models compared

I compared three classification models:

- Logistic Regression
- Random Forest
- Naive Bayes (additional method)

To evaluate performance, I used two metrics:

- Accuracy: percent of correct predictions
- AUC: how well the model separates High vs Low across thresholds using the ROC curve

Classification results

On the test data, the performance results were:

- Logistic Regression: AUC = 0.591, Accuracy = 0.558
- Naive Bayes: AUC = 0.571, Accuracy = 0.538
- Random Forest: AUC = 0.538, Accuracy = 0.540

Logistic regression performed the best overall, though none of the models were extremely strong since the AUC values were only slightly above 0.5. This suggests there is some signal in the predictors, but many important drivers of weekly sales are not present in the dataset.

Model <chr>	AUC <dbl>	Accuracy <dbl>
Logistic Regression	0.5911652	0.5583333
Random Forest	0.5383850	0.5400000
Naive Bayes	0.5713643	0.5383333

3 rows

Table 1: Model comparison table (Accuracy + AUC)

```
Call:
glm(formula = formula_clf, family = binomial, data = train_data)

Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)   -1.228e+02  2.561e+02  -0.479   0.6318
Store         -3.316e-03  4.527e-03  -0.733   0.4638
Size           8.061e-06  9.604e-07   8.394   <2e-16 ***
Temperature   -6.876e-03  3.350e-03  -2.053   0.0401 *
Fuel_Price    -1.532e-01  2.189e-01  -0.700   0.4839
CPI           -1.773e-03  1.659e-03  -1.069   0.2850
Unemployment  -5.933e-03  3.461e-02  -0.171   0.8639
IsHolidayTRUE  1.394e-01  2.096e-01   0.665   0.5061
Year           6.108e-02  1.277e-01   0.478   0.6323
Week           6.360e-03  4.291e-03   1.482   0.1383
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1940.8  on 1399  degrees of freedom
Residual deviance: 1850.5  on 1390  degrees of freedom
AIC: 1870.5

Number of Fisher Scoring iterations: 4

         Actual
Predicted Low High
     Low  152  119
     High 146  183
```

```
Setting direction: controls < cases
```

```
Area under the curve: 0.5912
[1] 0.5583333
```

Figure 4a: Logistic Regression ROC/AUC Results

This figure shows the logistic regression classification performance on the test set. The confusion matrix compares predicted vs actual High/Low sales weeks. The model achieved an AUC of 0.5912 and an accuracy of 0.5583, which was the best overall performance among the three models.

```
         Actual
Predicted Low High
     Low  160  138
     High 138  164
```

```
Setting direction: controls < cases
```

```
Area under the curve: 0.5384
[1] 0.54
```

Figure 4b: Random Forest ROC/AUC Results

This figure shows the random forest classification performance on the test set, including the confusion matrix and ROC/AUC output. The model achieved an AUC of 0.5384 and an accuracy of 0.5400. Compared to the other models, the random forest had the lowest AUC, meaning it separated High vs Low weeks the worst in this setup.

```
         Actual
Predicted Low High
     Low  134  113
     High 164  189
```

```
Setting direction: controls < cases
```

```
Area under the curve: 0.5714
[1] 0.5383333
```

Figure 4c: Naive Bayes ROC/AUC Results

This figure shows the naive Bayes classification performance on the test set, including the confusion matrix and ROC/AUC output. The model achieved an AUC of 0.5714 and an accuracy of 0.5383. This model performed better than random forest in AUC but slightly worse than logistic regression overall.

Interpretation of classification results

From the logistic regression results, Store Size was one of the strongest predictors and it was statistically significant, which makes sense because larger stores typically sell more. Temperature also showed a small statistically significant negative effect, which could be capturing seasonal patterns indirectly or regional differences. Random Forest may have performed worse because the predictors used might not contain strong nonlinear interactions for the high vs low label, and because the balanced sample size (2,000) is small compared to the full dataset. Also, the high vs low label is a simplified target, and truly high sales weeks may depend on missing variables such as promotions.

Limitations

This dataset does not include some major factors that likely affect sales (promotions, advertising, competitor prices, local events, and detailed customer demographics). This is probably why the regression model had a low R-squared (~0.062). Weekly totals can also hide day-to-day patterns inside a week. Finally, the classification target (above vs below median) is useful for a simple classification task, but it can be too coarse because it treats slightly above median and extremely high sales the same way.

Conclusion

In this project, I analyzed Walmart weekly sales data to understand patterns and relationships. The main conclusions were:

- There is strong seasonality: sales are highest in November and December and lowest in January.
- Holiday weeks have significantly higher average weekly sales than non-holiday weeks (very small p-value from the t-test).
- Unemployment has a very weak relationship with weekly sales (correlation near zero), and store size, month, and holiday effects matter more.
- For prediction, logistic regression performed best (AUC about 0.591), but overall classification performance was only moderate, suggesting important predictors are missing.

Overall, the dataset supports that time of year, holidays, and store characteristics are the main drivers of weekly sales in this dataset, while unemployment is not a strong predictor.

References

- Walmart weekly sales dataset (Kaggle), merged train/stores/features files.